



Proyecto fin de máster

Optimización de modelos de lenguaje compactos para la síntesis de programas orientada a la resolución de ARC-AGI-1

Asier Marin Fernandez

Director: Fernando Antonio Rufo Jimenez
Valencia, 22 de julio de 2026

Motivación

Large Language Models Are Not Strong Abstract Reasoners

Gaël Gendron
University of Auckland
ggen187@aucklanduni.ac.nz

Qiming Bao
University of Auckland
qbao775@aucklanduni.ac.nz

Michael Witbrock
University of Auckland
m.witbrock@auckland.ac.nz

Gillian Dobbie
University of Auckland
g.dobbie@auckland.ac.nz

Motivación II

- Transformar escalado por fuerza bruta a la optimización inteligente de parámetros.
- Evitar la sobre-generalización y fuerza al sistema a concentrarse exclusivamente en descubrir transformaciones algorítmicas.
- Apostar por paradigmas que han demostrado eficiencia y eficacia para la resolución de problemas ARC-AGI.
- Necesidad técnica y estrategias científicas prometedoras en el ámbito del razonamiento complejo sin la saturación de recursos de modelos a gran escala.

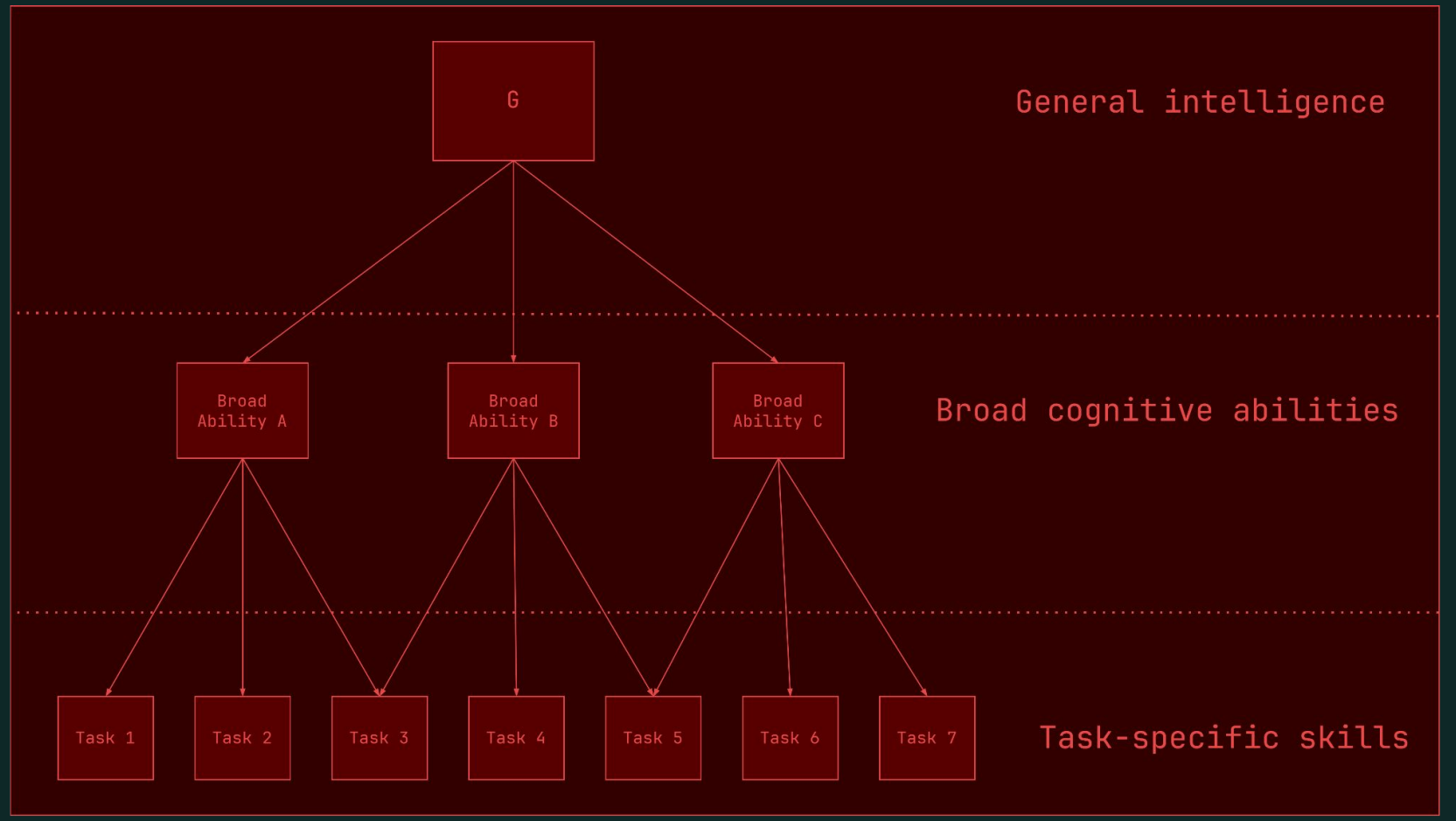
Objetivas

- Exploración técnicas de bucles de refinamiento.
- Desarrollar técnicas de entrenamiento en tiempo de inferencia.
- Optimización de consumo de memoria y tiempo de ejecución.
- Comparativa exhaustiva de eficiencia de manera cuantitativa respecto al uso de memoria VRAM y precisión teórica total en FLOPs.

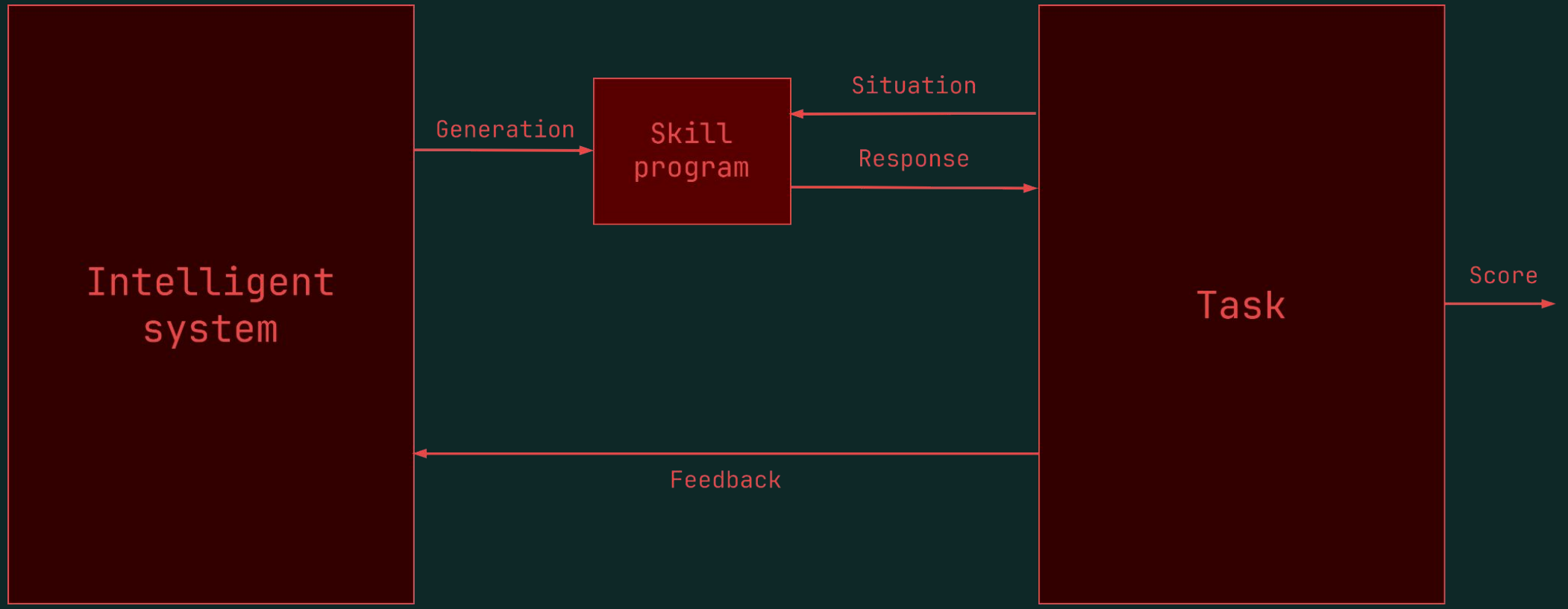
Alcance



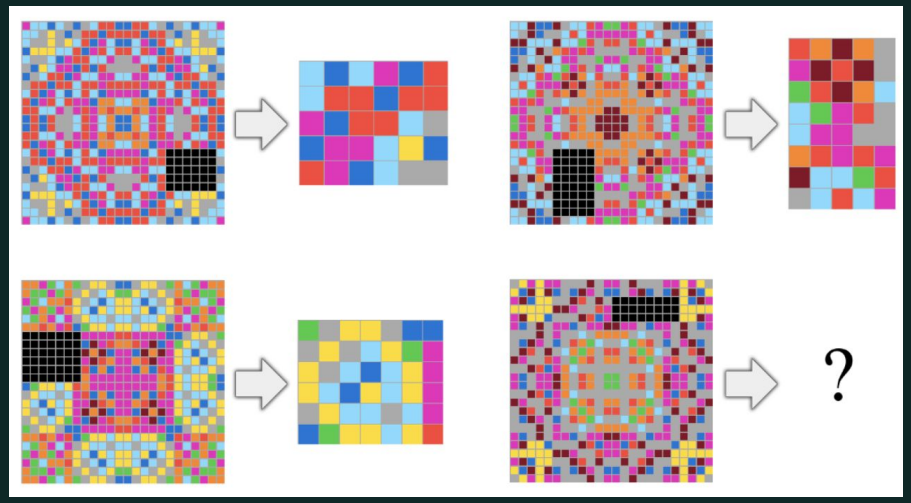
Contexto I

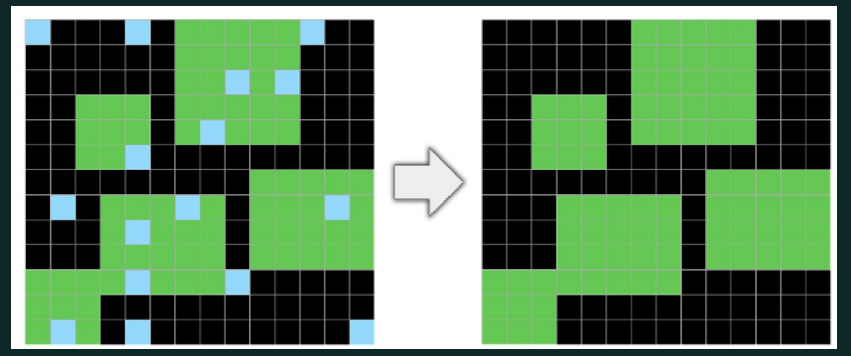

G
General intelligence
Broad Ability A
Broad Ability B
Broad Ability C
Broad cognitive abilities
Task 1
Task 2
Task 3
Task 4
Task 5
Task 6
Task 7
Task-specific skills

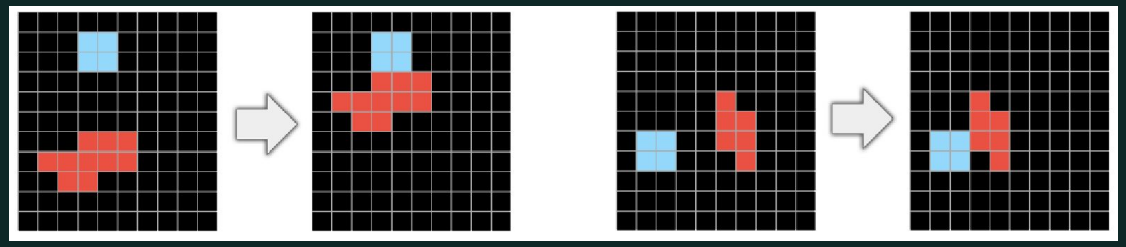
Contexto II

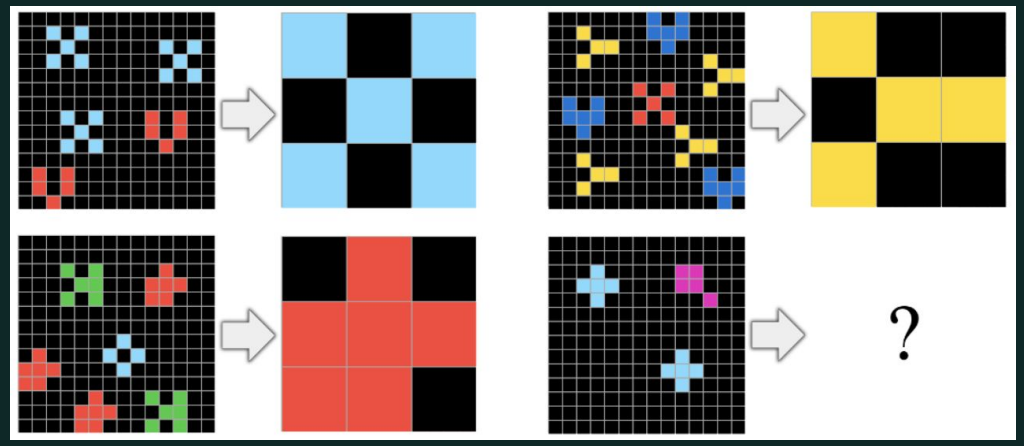

Intelligent system
Generation
Skill program
Situation
Response
Task
Score
Feedback

Contexto III

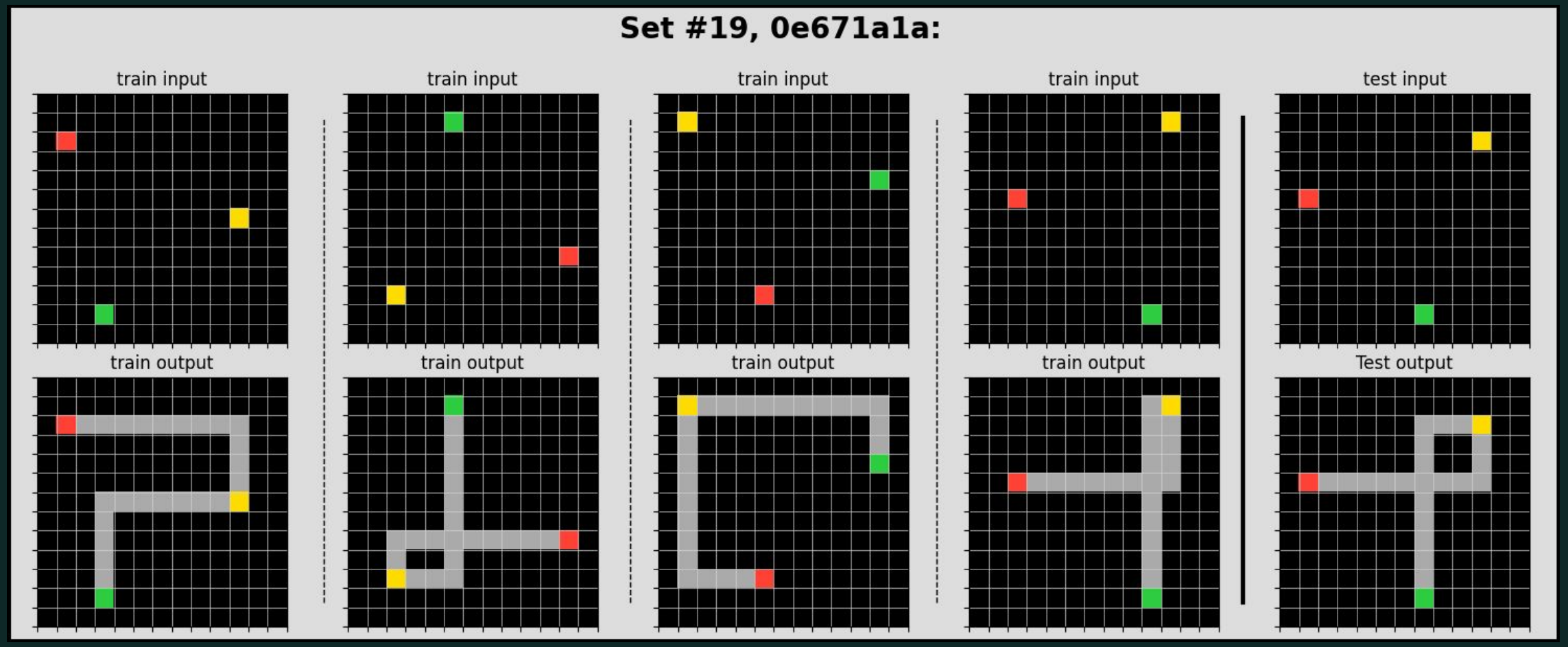


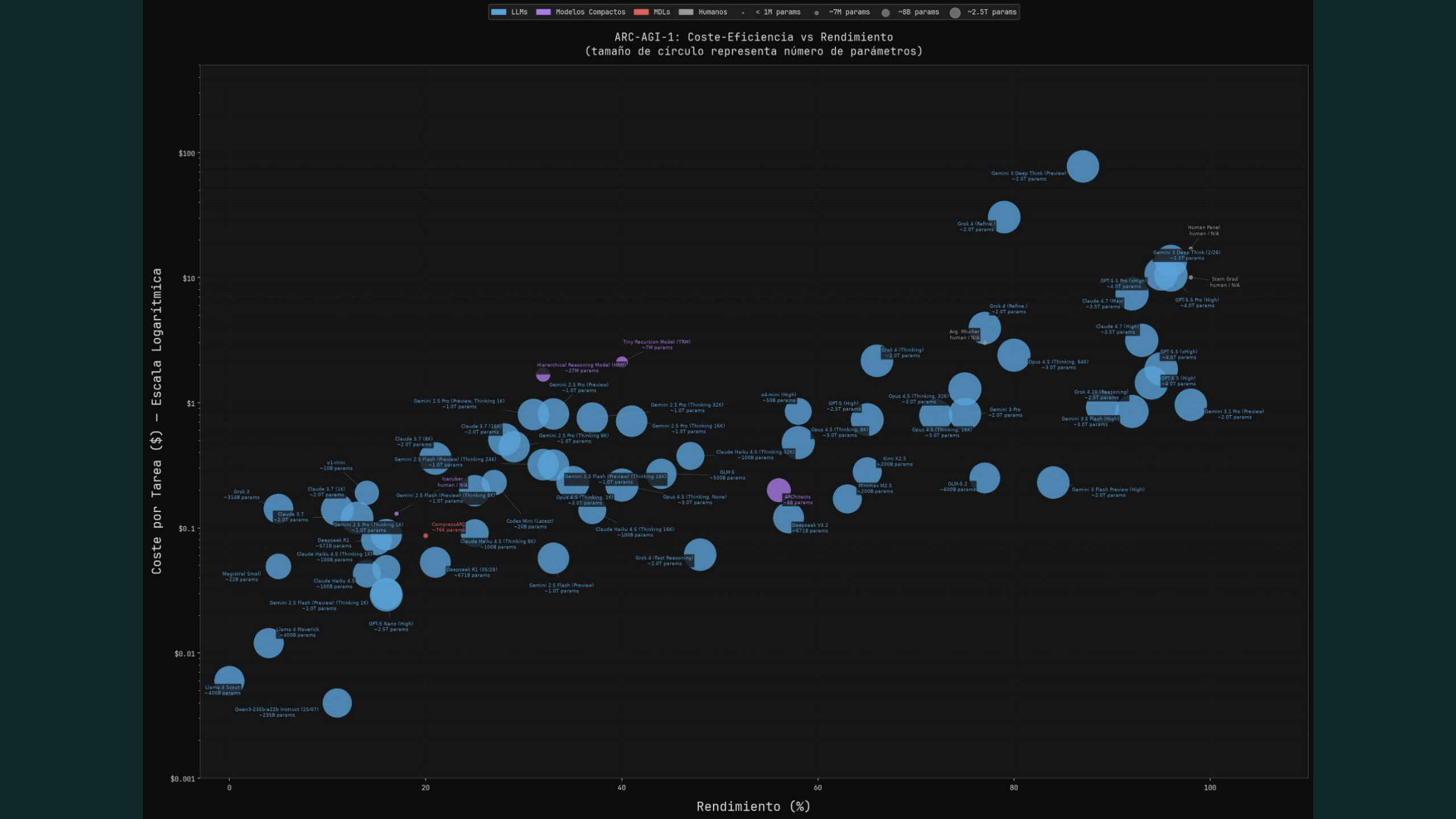


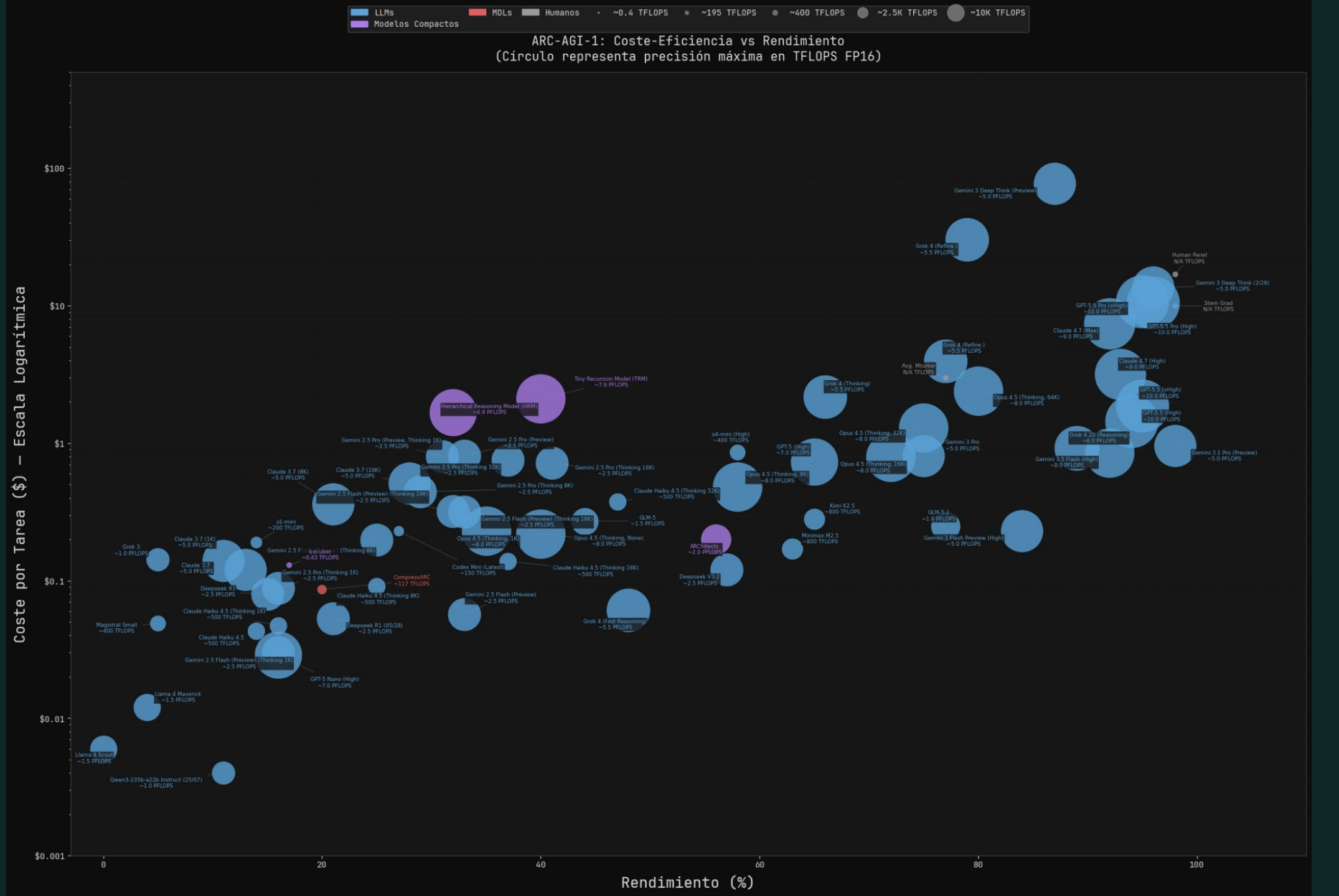


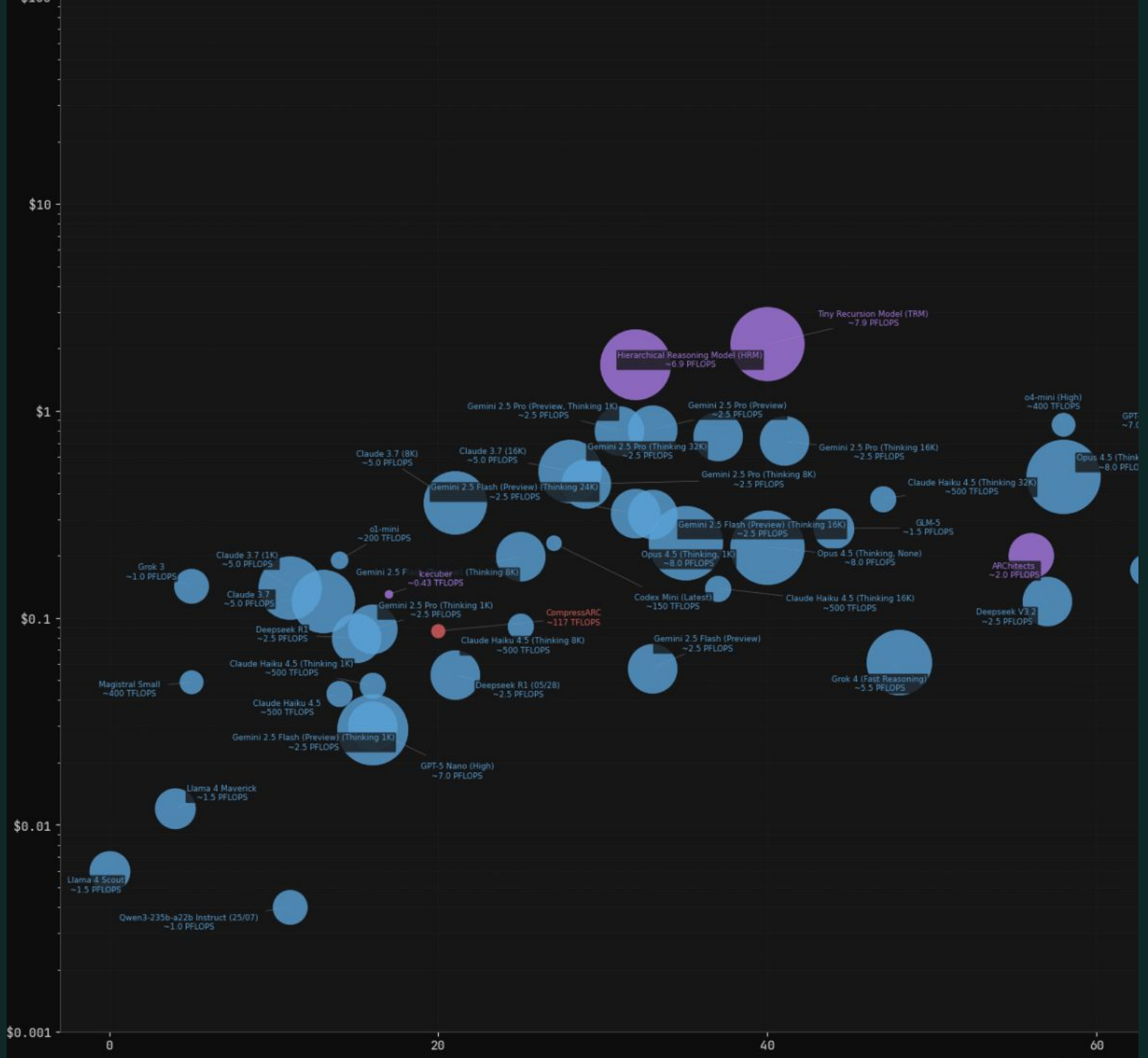


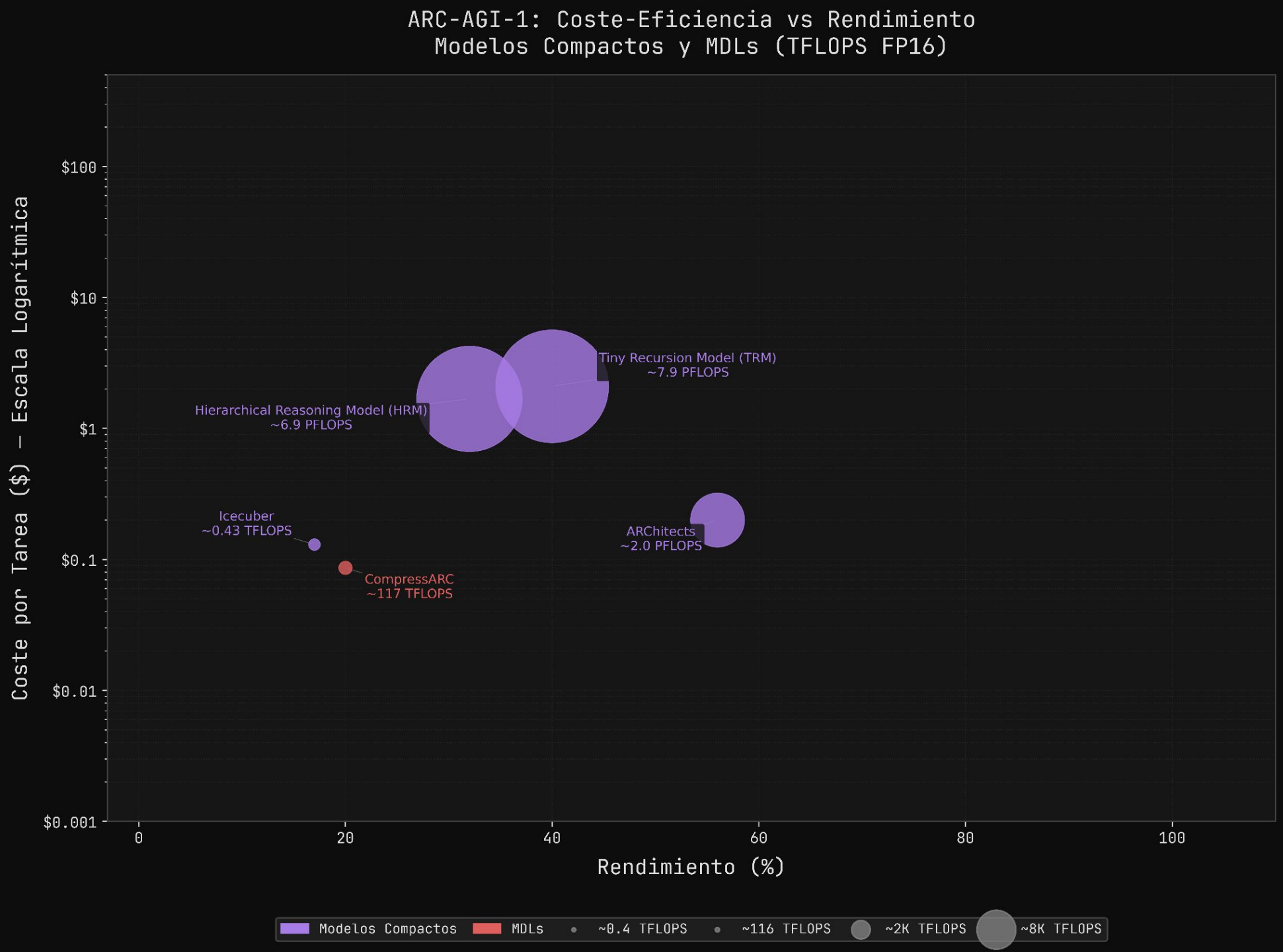
Contexto IV

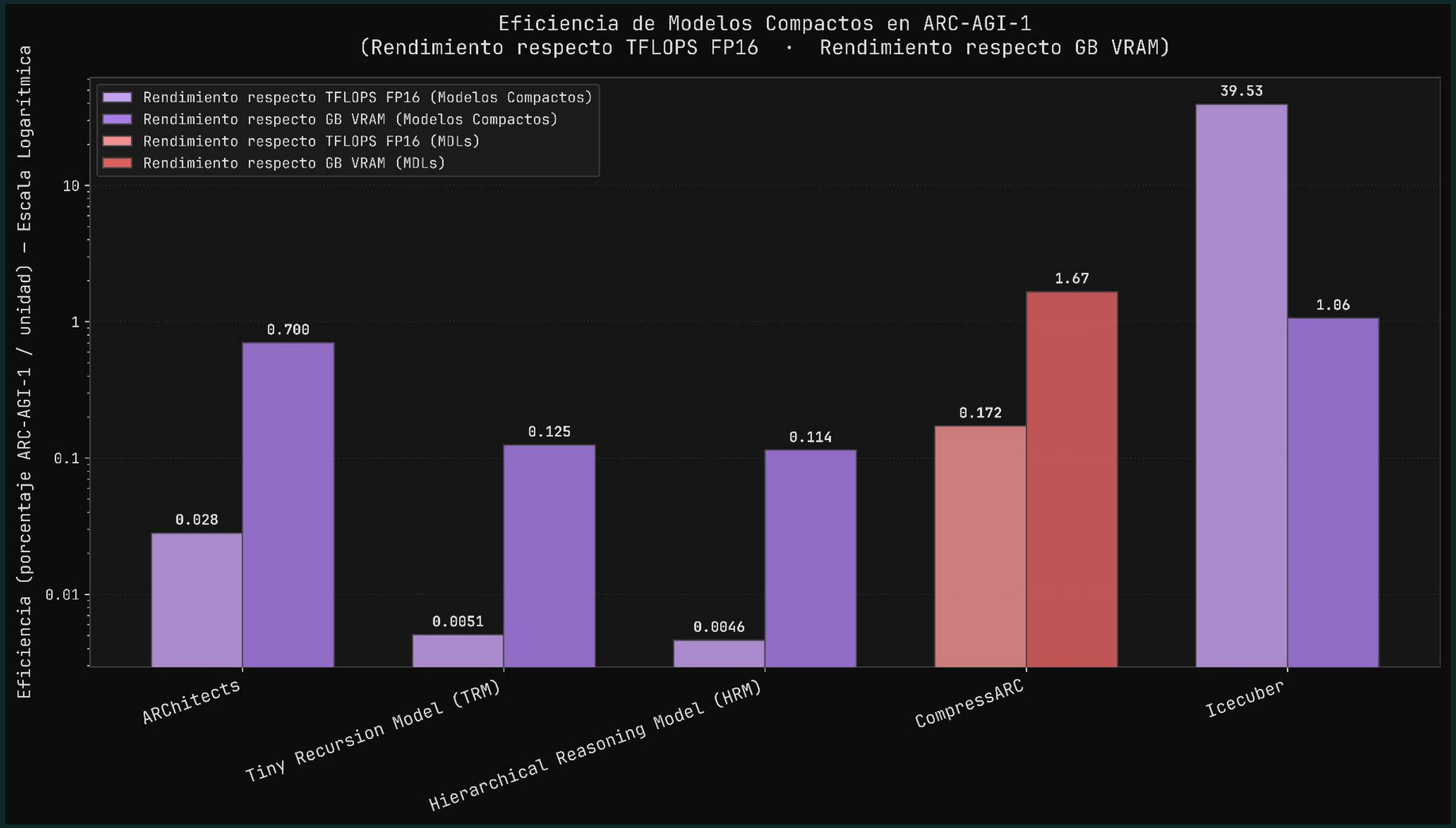

Set #19, 0e671a1a:
train input
train input
train input
train input
test input
train output
train output
train output
train output
Test output


LLMs
Modelos Compactos
MDLs
Humanos
< 1M params
~7M params
~8B params
~2.5T params
ARC-AGI-1: Coste-Eficiencia vs Rendimiento
(tamaño de círculo representa número de parámetros)
Coste por Tarea ($) — Escala Logarítmica
$100
$10
$1
$0.1
$0.01
$0.001
0
20
40
60
80
100
Rendimiento (%)
Gemini 3 Deep Think (Preview)
~2.0T params
Grok 4 (Refine.)
~2.0T params
Human Panel
human / N/A
Gemini 3 Deep Think (2/26)
~2.0T params
Stem Grad
human / N/A
GPT-5.5 Pro (xHigh)
~4.0T params
Claude 4.7 (Max)
~3.5T params
GPT-5.5 Pro (High)
~4.0T params
Grok 4 (Refine.)
~2.0T params
Avg. Mturker
human / N/A
Claude 4.7 (High)
~3.5T params
Grok 4 (Thinking)
~2.0T params
Opus 4.5 (Thinking, 64K)
~3.0T params
GPT-5.5 (xHigh)
~4.0T params
GPT-5.5 (High)
~4.0T params
Tiny Recursion Model (TRM)
~7M params
Hierarchical Reasoning Model (HRM)
~27M params
Gemini 2.5 Pro (Preview)
~1.0T params
o4-mini (High)
~50B params
GPT-5 (High)
~2.5T params
Opus 4.5 (Thinking, 32K)
~3.0T params
Gemini 3 Pro
~2.0T params
Opus 4.5 (Thinking, 16K)
~3.0T params
Grok 4.20 (Reasoning)
~2.5T params
Gemini 3.5 Flash (High)
~3.0T params
Gemini 3.1 Pro (Preview)
~2.0T params
Gemini 2.5 Pro (Preview, Thinking 1K)
~1.0T params
Gemini 2.5 Pro (Thinking 32K)
~1.0T params
Claude 3.7 (16K)
~2.0T params
Gemini 2.5 Pro (Thinking 16K)
~1.0T params
Gemini 2.5 Pro (Thinking 8K)
~1.0T params
Claude 3.7 (8K)
~2.0T params
Opus 4.5 (Thinking, 8K)
~3.0T params
Claude Haiku 4.5 (Thinking 32K)
~100B params
Gemini 2.5 Flash (Preview) (Thinking 24K)
~1.0T params
o1-mini
~100B params
Kimi K2.5
~200B params
Gemini 2.5 Flash (Preview) (Thinking 16K)
~1.0T params
GLM-5
~500B params
Icecuber
human / N/A
Grok 3
~314B params
Claude 3.7 (1K)
~2.0T params
Gemini 2.5 Flash (Preview) (Thinking 8K)
~1.0T params
Opus 4.5 (Thinking, None)
~3.0T params
Minimax M2.5
~200B params
ARChitects
~8B params
GLM-5.2
~600B params
Gemini 3 Flash Preview (High)
~2.0T params
Claude 3.7
~2.0T params
Opus 4.5 (Thinking, 1K)
~3.0T params
Codex Mini (Latest)
~20B params
Gemini 2.5 Pro (Thinking 1K)
~1.0T params
CompressARC
~76K params
Claude Haiku 4.5 (Thinking 16K)
~100B params
Deepseek V3.2
~671B params
Deepseek R1
~671B params
Claude Haiku 4.5 (Thinking 8K)
~100B params
Claude Haiku 4.5 (Thinking 1K)
~100B params
Grok 4 (Fast Reasoning)
~2.0T params
Magistral Small
~22B params
Deepseek R1 (05/28)
~671B params
Claude Haiku 4.5
~100B params
Gemini 2.5 Flash (Preview)
~1.0T params
Gemini 2.5 Flash (Preview) (Thinking 1K)
~1.0T params
GPT-5 Nano (High)
~2.5T params
Llama 4 Maverick
~400B params
Llama 4 Scout
~400B params
Qwen3-235b-a22b Instruct (25/07)
~235B params


LLMs
Modelos Compactos
MDLs
Humanos
~0.4 TFLOPS
~195 TFLOPS
~400 TFLOPS
~2.5K TFLOPS
~10K TFLOPS
ARC-AGI-1: Coste-Eficiencia vs Rendimiento
(Círculo representa precisión máxima en TFLOPS FP16)
Coste por Tarea ($) — Escala Logarítmica
$100
$10
$1
$0.1
$0.01
$0.001
0
20
40
60
80
100
Rendimiento (%)
Gemini 3 Deep Think (Preview) ~5.0 PFLOPS
Grok 4 (Refine.) ~5.5 PFLOPS
Human Panel N/A TFLOPS
Gemini 3 Deep Think (2/26) ~5.0 PFLOPS
Stem Grad N/A TFLOPS
GPT-5.5 Pro (xHigh) ~10.0 PFLOPS
GPT-5.5 Pro (High) ~10.0 PFLOPS
Claude 4.7 (Max) ~9.0 PFLOPS
Grok 4 (Refine.) ~5.5 PFLOPS
Avg. Mturker N/A TFLOPS
Claude 4.7 (High) ~9.0 PFLOPS
GPT-5.5 (xHigh) ~10.0 PFLOPS
GPT-5.5 (High) ~10.0 PFLOPS
Grok 4.20 (Reasoning) ~6.0 PFLOPS
Gemini 3.1 Pro (Preview) ~5.0 PFLOPS
Gemini 3.5 Flash (High) ~8.0 PFLOPS
Tiny Recursion Model (TRM) ~7.9 PFLOPS
Hierarchical Reasoning Model (HRM) ~6.9 PFLOPS
Grok 4 (Thinking) ~5.5 PFLOPS
Opus 4.5 (Thinking, 64K) ~8.0 PFLOPS
Opus 4.5 (Thinking, 32K) ~8.0 PFLOPS
Gemini 3 Pro ~5.0 PFLOPS
Opus 4.5 (Thinking, 16K) ~8.0 PFLOPS
GPT-5 (High) ~7.0 PFLOPS
o4-mini (High) ~400 TFLOPS
Opus 4.5 (Thinking, 8K) ~8.0 PFLOPS
Gemini 2.5 Pro (Preview, Thinking 1K) ~2.5 PFLOPS
Gemini 2.5 Pro (Preview) ~2.5 PFLOPS
Gemini 2.5 Pro (Thinking 32K) ~2.5 PFLOPS
Gemini 2.5 Pro (Thinking 16K) ~2.5 PFLOPS
Claude 3.7 (8K) ~5.0 PFLOPS
Claude 3.7 (16K) ~5.0 PFLOPS
Gemini 2.5 Pro (Thinking 8K) ~2.5 PFLOPS
Gemini 2.5 Flash (Preview) (Thinking 24K) ~2.5 PFLOPS
Claude Haiku 4.5 (Thinking 32K) ~500 TFLOPS
Kimi K2.5 ~800 TFLOPS
GLM-5.2 ~1.8 PFLOPS
Gemini 3 Flash Preview (High) ~5.0 PFLOPS
Minimax M2.5 ~800 TFLOPS
GLM-5 ~1.5 PFLOPS
Gemini 2.5 Flash (Preview) (Thinking 16K) ~2.5 PFLOPS
Opus 4.5 (Thinking, None) ~8.0 PFLOPS
Opus 4.5 (Thinking, 1K) ~8.0 PFLOPS
o1-mini ~200 TFLOPS
Claude 3.7 (1K) ~5.0 PFLOPS
Grok 3 ~1.0 PFLOPS
Gemini 2.5 F
Icecuber ~0.43 TFLOPS
(Thinking 8K)
Claude 3.7 ~5.0 PFLOPS
Gemini 2.5 Pro (Thinking 1K) ~2.5 PFLOPS
Codex Mini (Latest) ~150 TFLOPS
Claude Haiku 4.5 (Thinking 16K) ~500 TFLOPS
ARChitects ~2.0 PFLOPS
Deepseek V3.2 ~2.5 PFLOPS
CompressARC ~117 TFLOPS
Deepseek R1 ~2.5 PFLOPS
Claude Haiku 4.5 (Thinking 8K) ~500 TFLOPS
Gemini 2.5 Flash (Preview) ~2.5 PFLOPS
Claude Haiku 4.5 (Thinking 1K) ~500 TFLOPS
Magistral Small ~400 TFLOPS
Deepseek R1 (05/28) ~2.5 PFLOPS
Grok 4 (Fast Reasoning) ~5.5 PFLOPS
Claude Haiku 4.5 ~500 TFLOPS
Gemini 2.5 Flash (Preview) (Thinking 1K) ~2.5 PFLOPS
GPT-5 Nano (High) ~7.0 PFLOPS
Llama 4 Maverick ~1.5 PFLOPS
Llama 4 Scout ~1.5 PFLOPS
Qwen3-235b-a22b Instruct (25/07) ~1.0 PFLOPS


$10
$1
$0.1
$0.01
$0.001
0
20
40
60
Tiny Recursion Model (TRM)
~7.9 PFLOPS
Hierarchical Reasoning Model (HRM)
~6.9 PFLOPS
Gemini 2.5 Pro (Preview, Thinking 1K)
~2.5 PFLOPS
Gemini 2.5 Pro (Preview)
~2.5 PFLOPS
Gemini 2.5 Pro (Thinking 32K)
~2.5 PFLOPS
Gemini 2.5 Pro (Thinking 16K)
~2.5 PFLOPS
Gemini 2.5 Pro (Thinking 8K)
~2.5 PFLOPS
o4-mini (High)
~400 TFLOPS
Claude 3.7 (8K)
~5.0 PFLOPS
Claude 3.7 (16K)
~5.0 PFLOPS
Gemini 2.5 Flash (Preview) (Thinking 24K)
~2.5 PFLOPS
Claude Haiku 4.5 (Thinking 32K)
~500 TFLOPS
Gemini 2.5 Flash (Preview) (Thinking 16K)
~2.5 PFLOPS
GLM-5
~1.5 PFLOPS
o1-mini
~200 TFLOPS
Claude 3.7 (1K)
~5.0 PFLOPS
Grok 3
~1.0 PFLOPS
Gemini 2.5 F
Icecuber
~0.43 TFLOPS
(Thinking 8K)
Opus 4.5 (Thinking, 1K)
~8.0 PFLOPS
Opus 4.5 (Thinking, None)
~8.0 PFLOPS
ARChitects
~2.0 PFLOPS
Claude 3.7
~5.0 PFLOPS
Gemini 2.5 Pro (Thinking 1K)
~2.5 PFLOPS
Codex Mini (Latest)
~150 TFLOPS
Claude Haiku 4.5 (Thinking 16K)
~500 TFLOPS
Deepseek V3.2
~2.5 PFLOPS
CompressARC
~117 TFLOPS
Deepseek R1
~2.5 PFLOPS
Claude Haiku 4.5 (Thinking 8K)
~500 TFLOPS
Gemini 2.5 Flash (Preview)
~2.5 PFLOPS
Claude Haiku 4.5 (Thinking 1K)
~500 TFLOPS
Magistral Small
~400 TFLOPS
Deepseek R1 (05/28)
~2.5 PFLOPS
Grok 4 (Fast Reasoning)
~3.5 PFLOPS
Claude Haiku 4.5
~500 TFLOPS
Gemini 2.5 Flash (Preview) (Thinking 1K)
~2.5 PFLOPS
GPT-5 Nano (High)
~7.0 PFLOPS
Llama 4 Maverick
~1.5 PFLOPS
Llama 4 Scout
~1.5 PFLOPS
Qwen3-235b-a22b Instruct (25/07)
~1.0 PFLOPS


ARC-AGI-1: Coste-Eficiencia vs Rendimiento
Modelos Compactos y MDLs (TFLOPS FP16)
Coste por Tarea ($) — Escala Logarítmica
$100
$10
$1
$0.1
$0.01
$0.001
0
20
40
60
80
100
Rendimiento (%)
Tiny Recursion Model (TRM)
~7.9 PFLOPS
Hierarchical Reasoning Model (HRM)
~6.9 PFLOPS
Icecuber
~0.43 TFLOPS
CompressARC
~117 TFLOPS
ARChitects
~2.0 PFLOPS
Modelos Compactos
MDLs
~0.4 TFLOPS
~116 TFLOPS
~2K TFLOPS
~8K TFLOPS

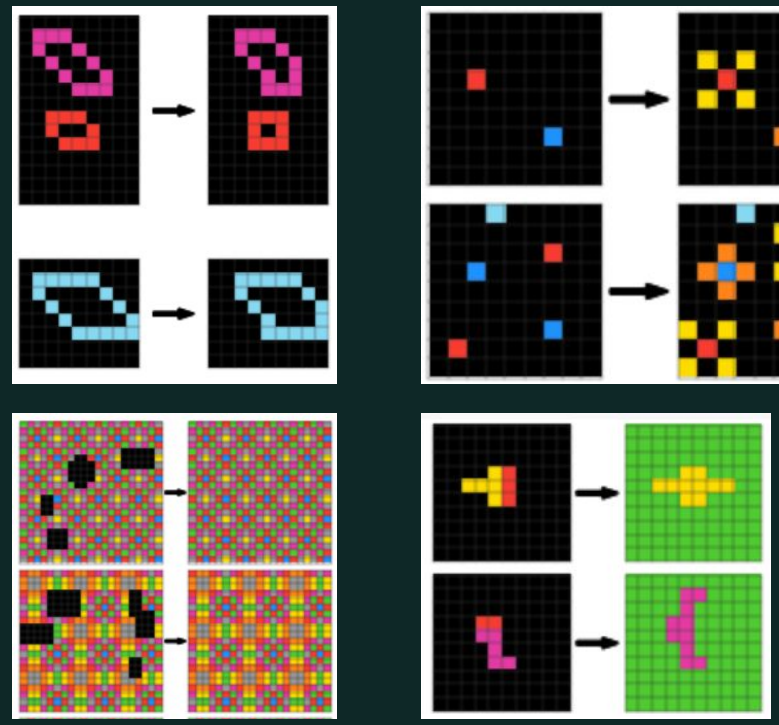

Eficiencia de Modelos Compactos en ARC-AGI-1
(Rendimiento respecto TFLOPS FP16 · Rendimiento respecto GB VRAM)
Eficiencia (porcentaje ARC-AGI-1 / unidad) — Escala Logarítmica
Rendimiento respecto TFLOPS FP16 (Modelos Compactos)
Rendimiento respecto GB VRAM (Modelos Compactos)
Rendimiento respecto TFLOPS FP16 (MDLs)
Rendimiento respecto GB VRAM (MDLs)
10
1
0.1
0.01
0.028
0.700
0.0051
0.125
0.0046
0.114
0.172
1.67
39.53
1.06
ARChitects
Tiny Recursion Model (TRM)
Hierarchical Reasoning Model (HRM)
CompressARC
Icecuber

CompressARC



CompressARC II

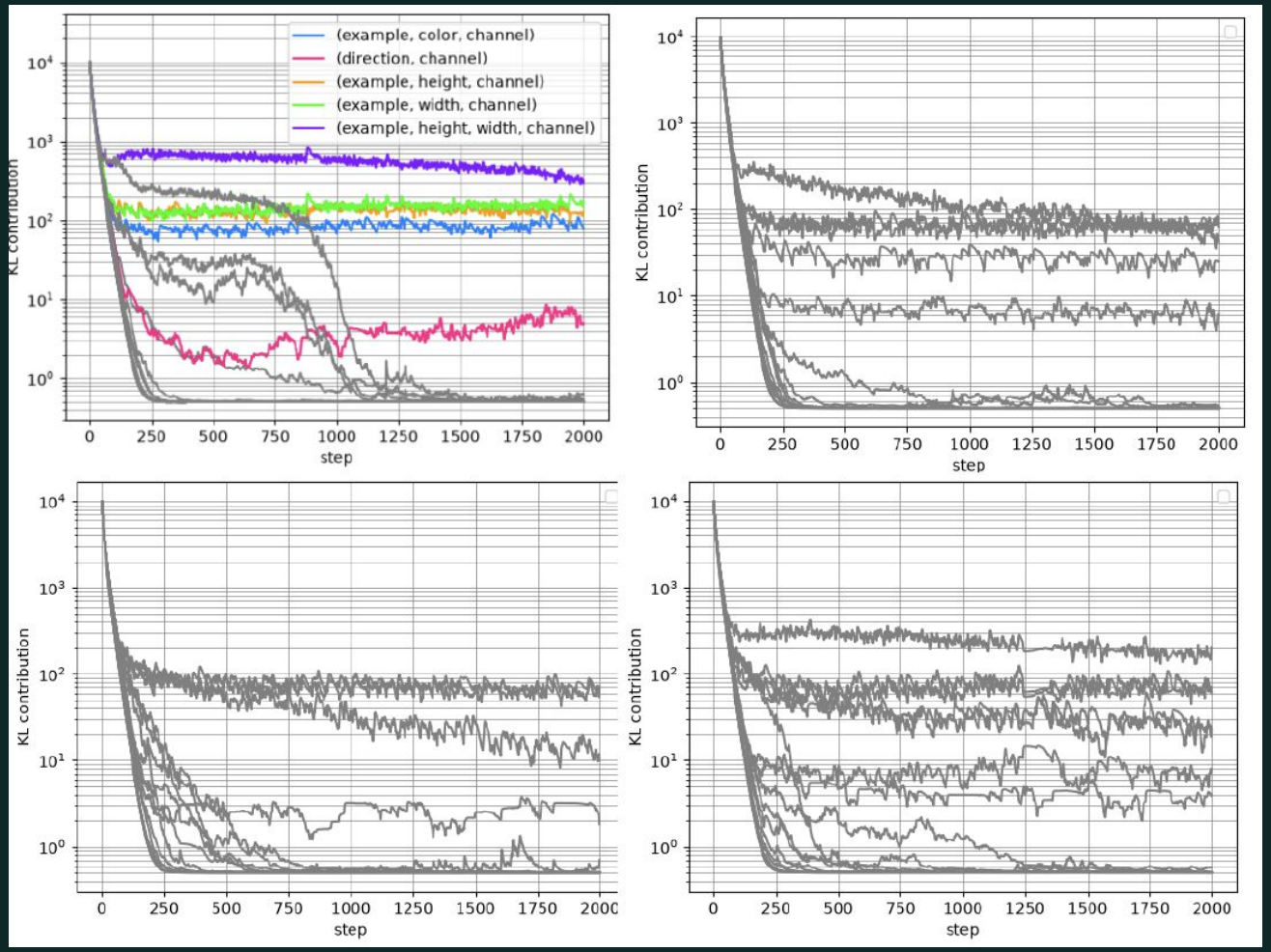
- Completar patrones espaciales
- LLenar regiones cerradas
- Detectar y completar simetrías: espejos, rotaciones y traslaciones
- Agrupar objetos
- Aplicar transformaciones simples a regiones
- Identificar el objeto distinto en una región uniforme



CompressARC III

- **Ausencia de habilidades primitivas como contar.**
- **Uso excesivo de CPU** con entrenamiento de tareas en paralelo.
- **Limitaciones que requieren recurrencia** en el tiempo como simulaciones de píxeles.
- **Ausencia de planificación** a largo plazo.
- **Aleatoriedad en la semilla**, algunas veces resuelve la misma tarea y otras no.
- **Limitaciones estructurales** de las capas densas (decodificación).
- **Colapso del espacio latente**, 4 tensores sostienen el aprendizaje dentro del bucle iterativo.

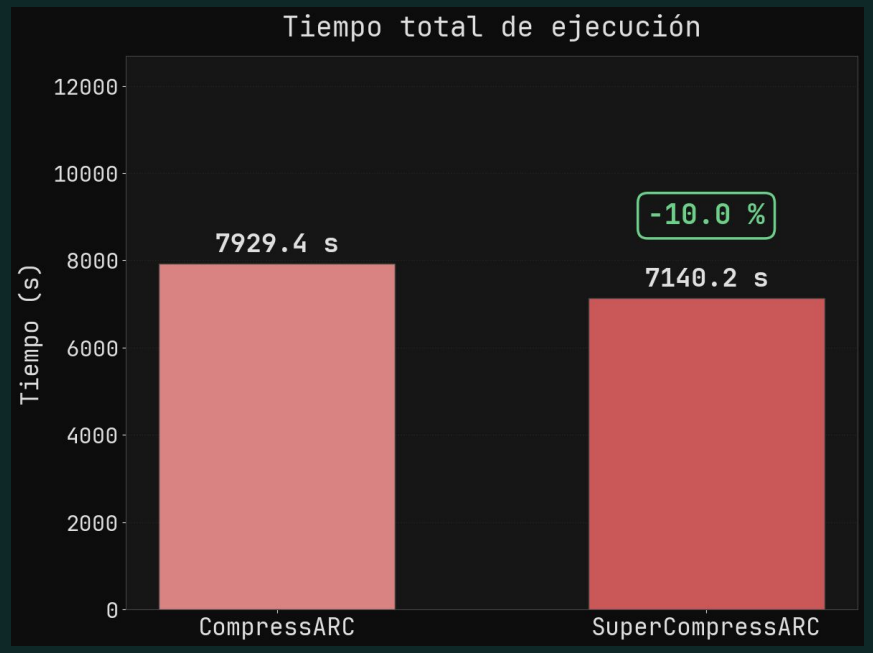
CompressARC IV

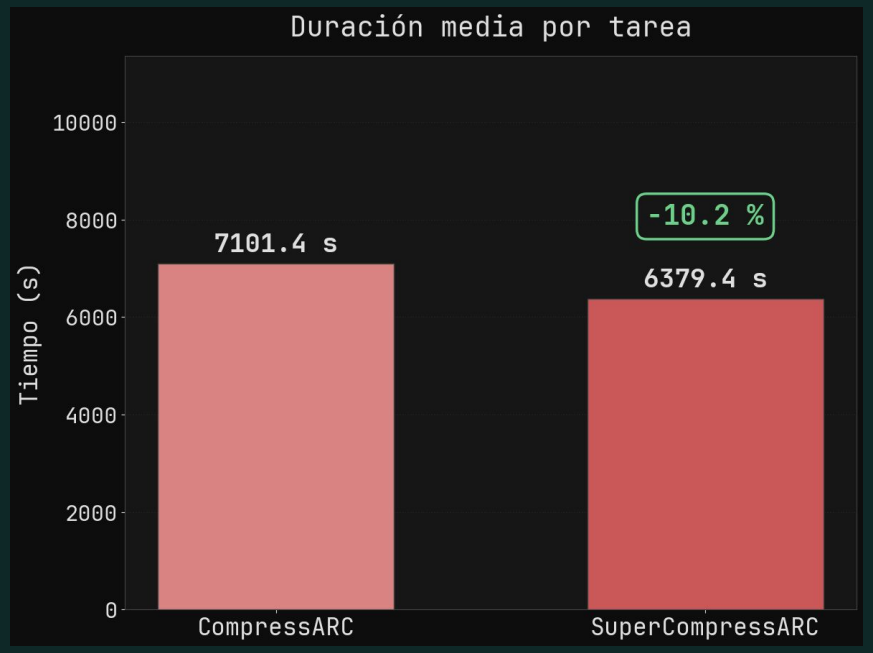


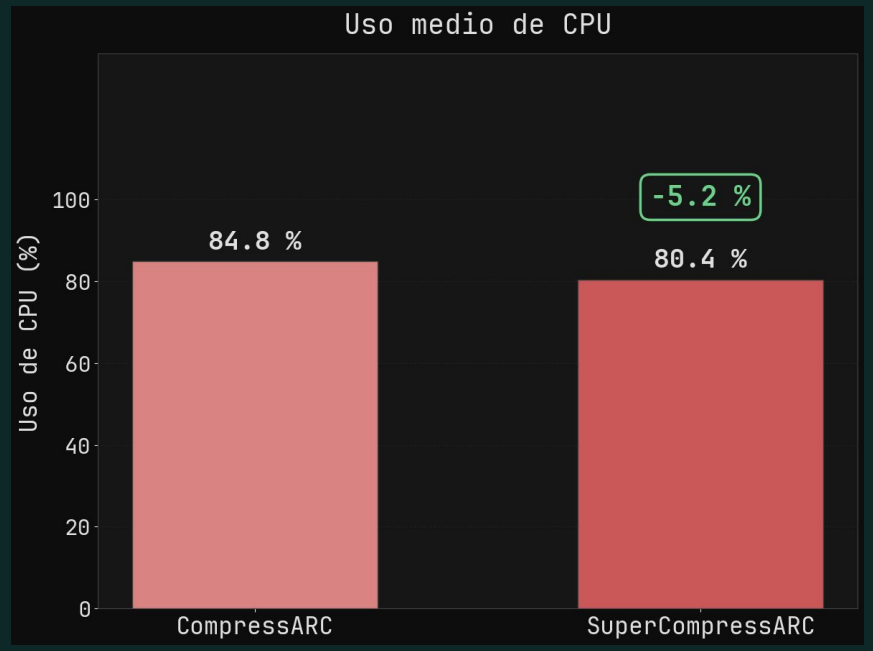
SuperCompressArc

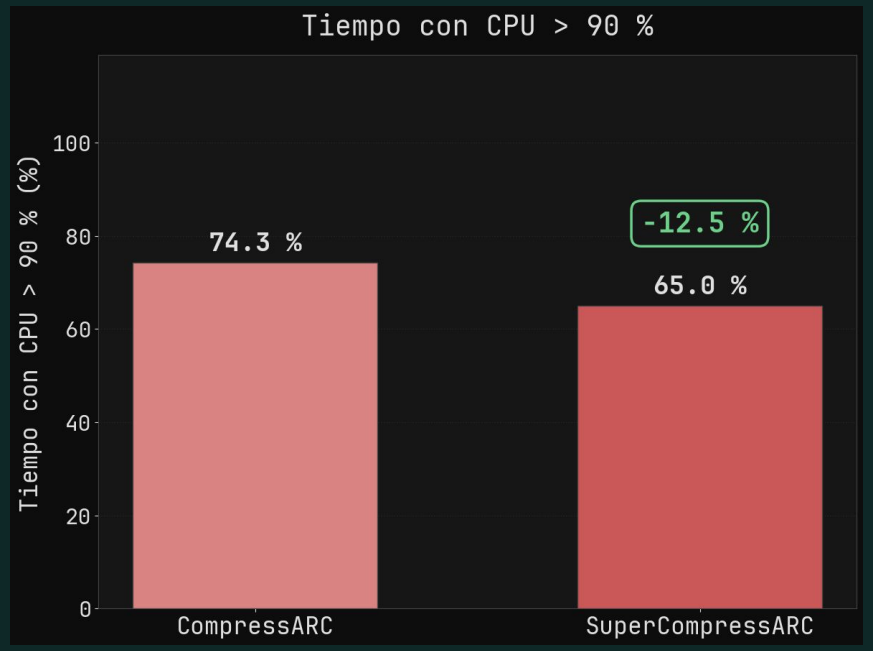
Eje H: Reducción de saturación de procesos en paralelo




Tiempo total de ejecución
Tiempo (s)
12000
10000
8000
6000
4000
2000
0
7929.4 s
7140.2 s
-10.0 %
CompressARC
SuperCompressARC


Duración media por tarea
Tiempo (s)
10000
8000
6000
4000
2000
0
7101.4 s
6379.4 s
-10.2 %
CompressARC
SuperCompressARC

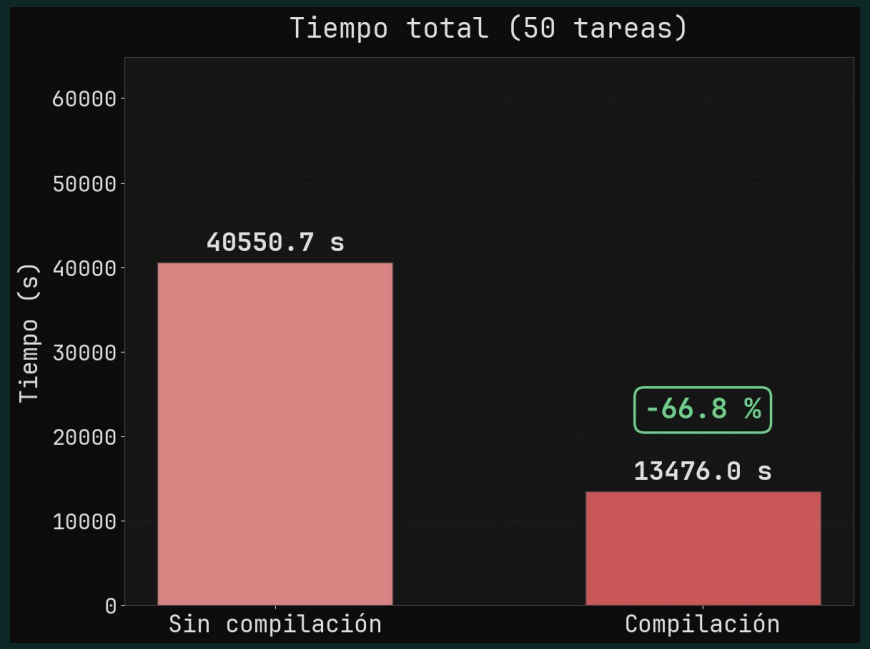

Uso medio de CPU
Uso de CPU (%)
100
80
60
40
20
0
84.8 %
80.4 %
-5.2 %
CompressARC
SuperCompressARC

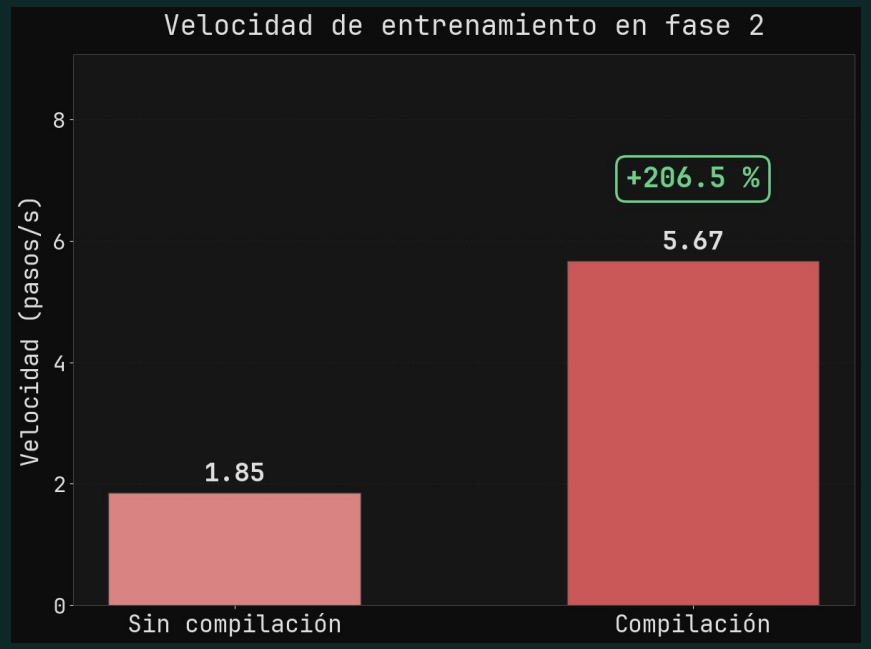

Tiempo con CPU > 90 %
Tiempo con CPU > 90 % (%)
100
80
60
40
20
0
74.3 %
65.0 %
-12.5 %
CompressARC
SuperCompressARC

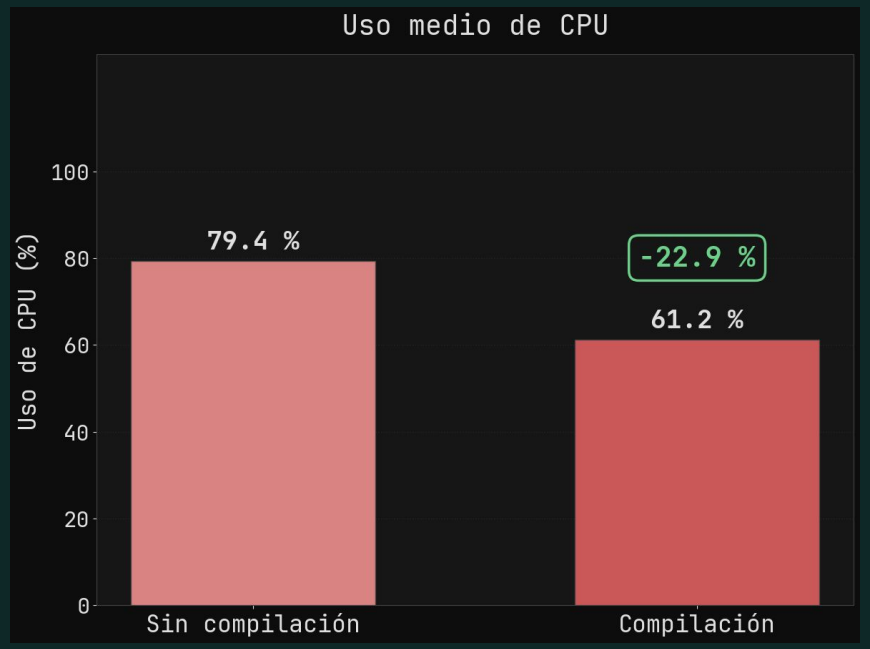
SuperCompressArc II

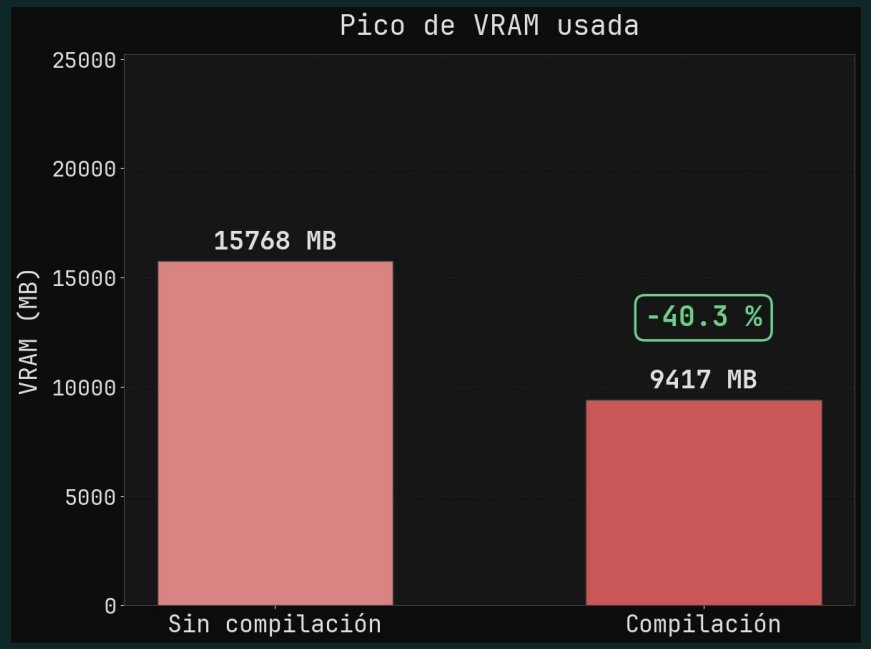
Eje H + D: Eficiencia computacional y aprovechamiento de recursos




Tiempo total (50 tareas)
Tiempo (s)
60000
50000
40000
30000
20000
10000
0
40550.7 s
-66.8 %
13476.0 s
Sin compilación
Compilación


Velocidad de entrenamiento en fase 2
Velocidad (pasos/s)
8
6
4
2
0
+206.5 %
5.67
1.85
Sin compilación
Compilación


Uso medio de CPU
Uso de CPU (%)
100
80
60
40
20
0
79.4 %
-22.9 %
61.2 %
Sin compilación
Compilación

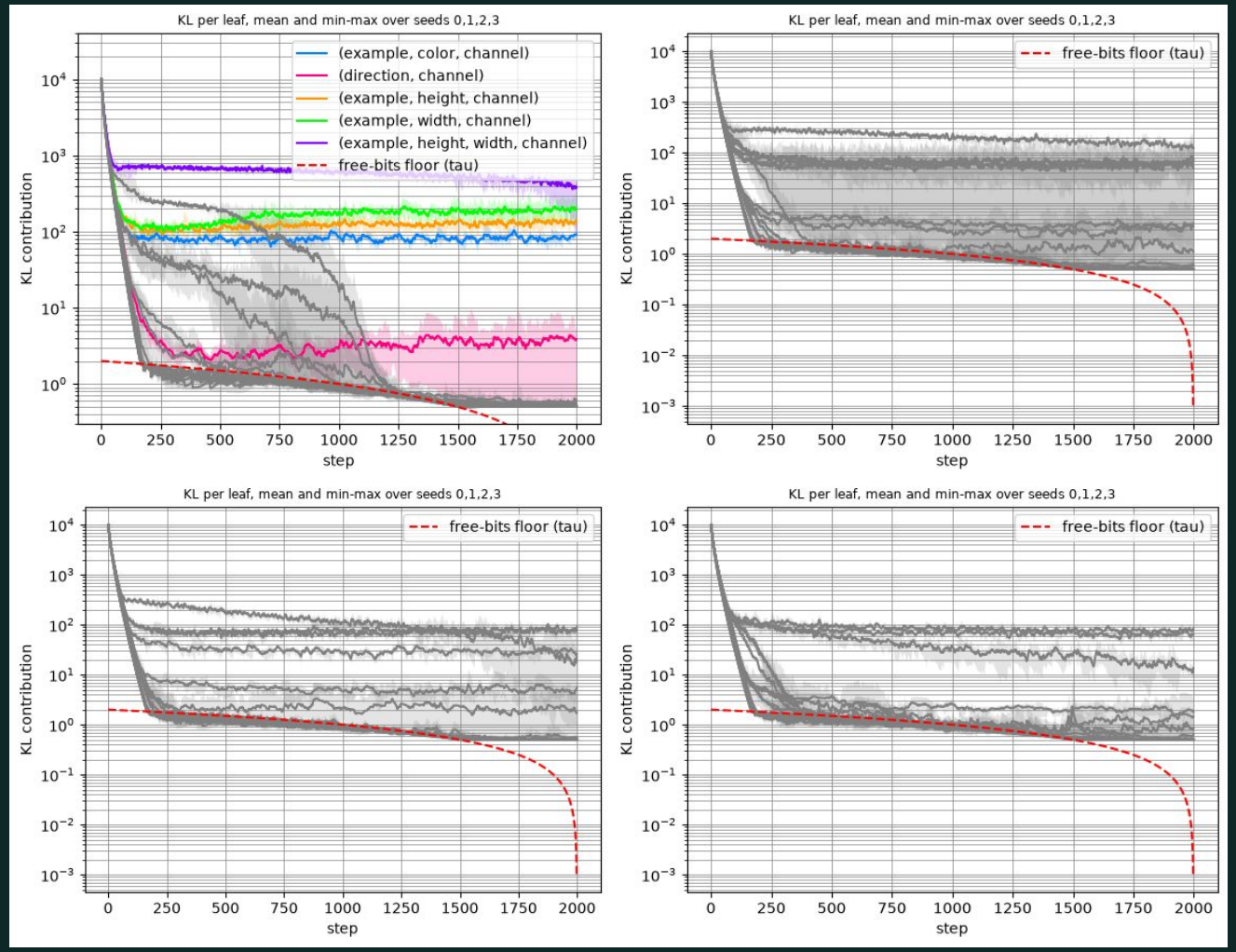

Pico de VRAM usada
VRAM (MB)
25000
20000
15000
10000
5000
0
15768 MB
-40.3 %
9417 MB
Sin compilación
Compilación

SuperCompressArc III

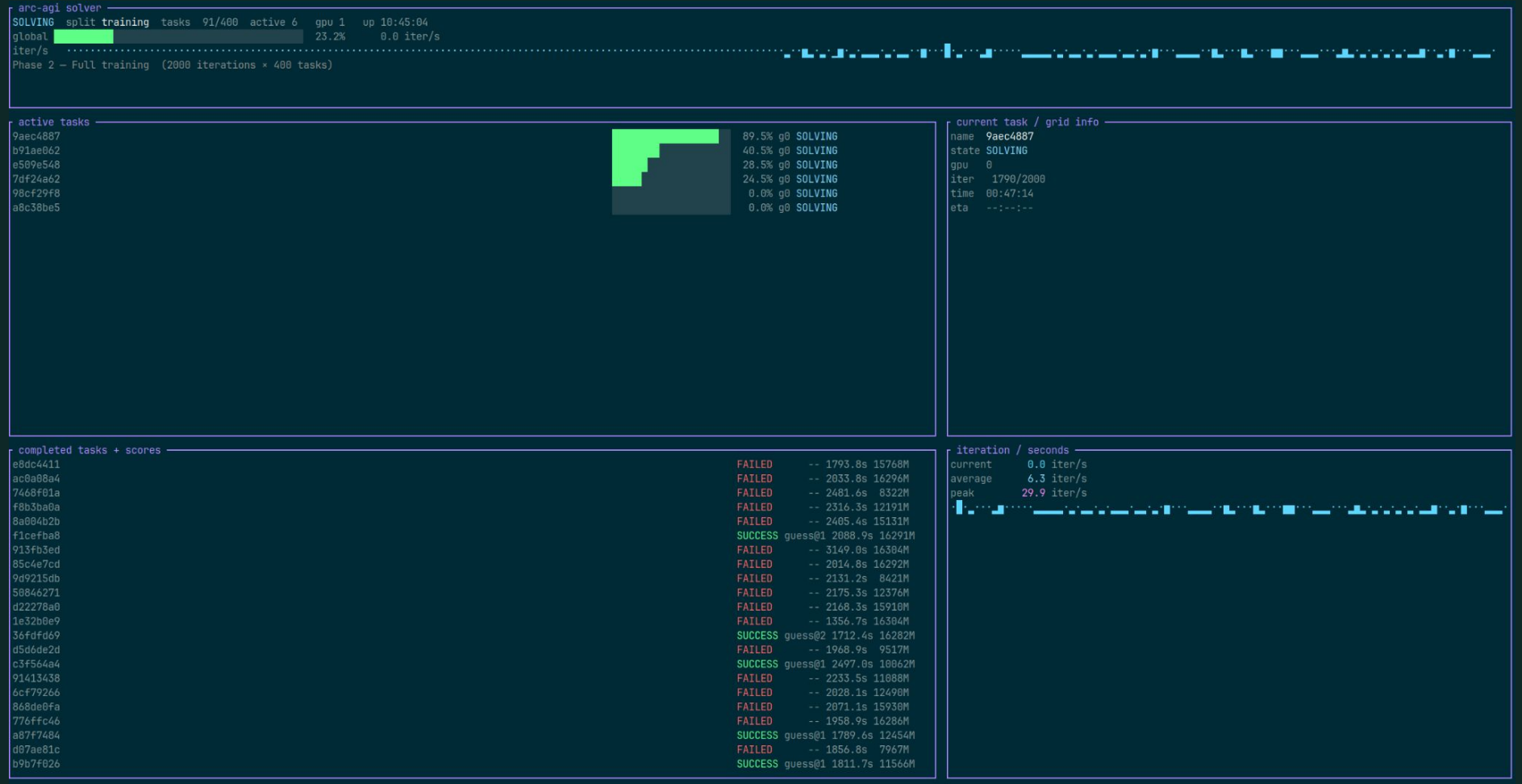
Eje H + D + B: Mejora de robustez del espacio latente



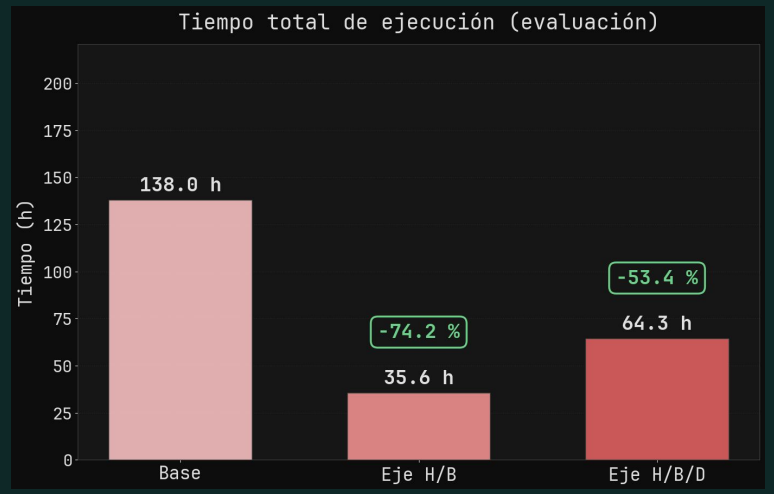
SuperCompressArc IV

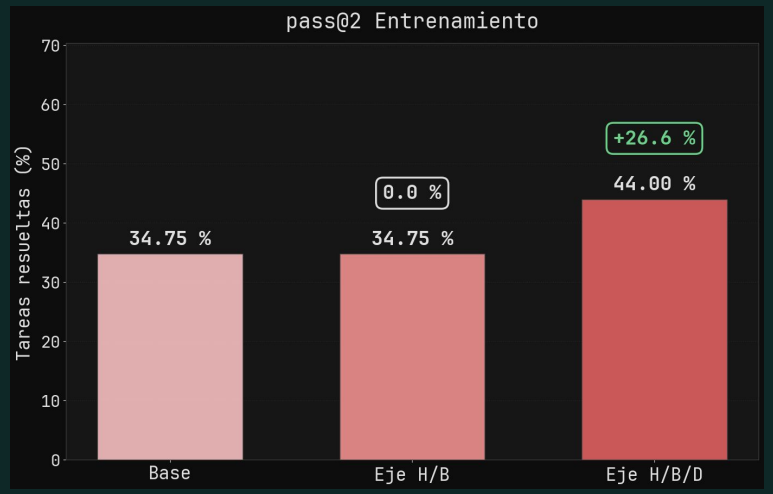


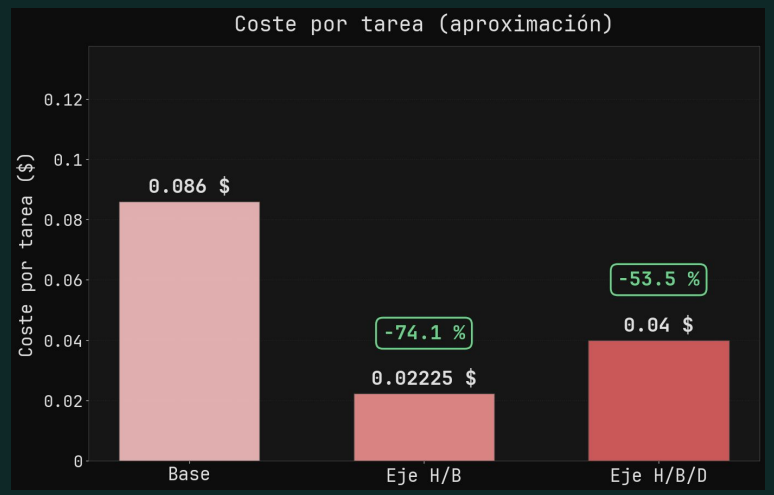
SuperCompressArc V

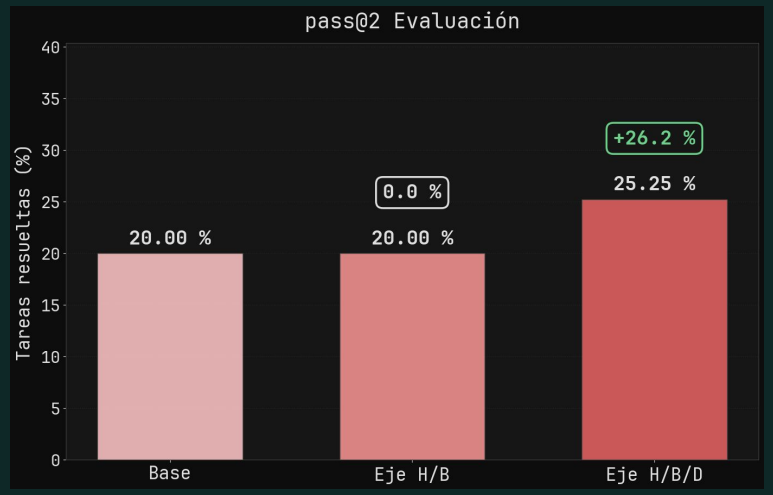


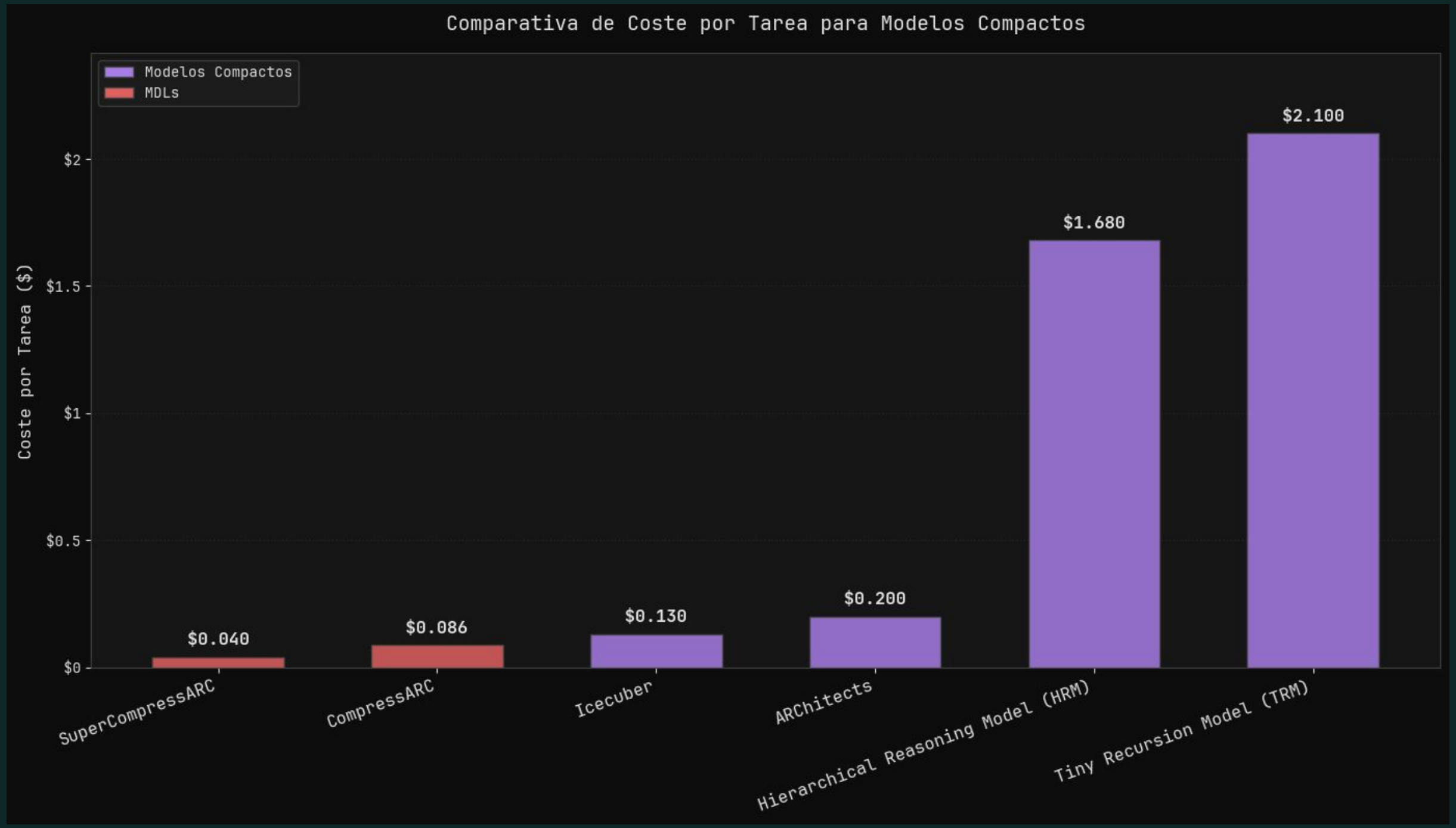
Resultados

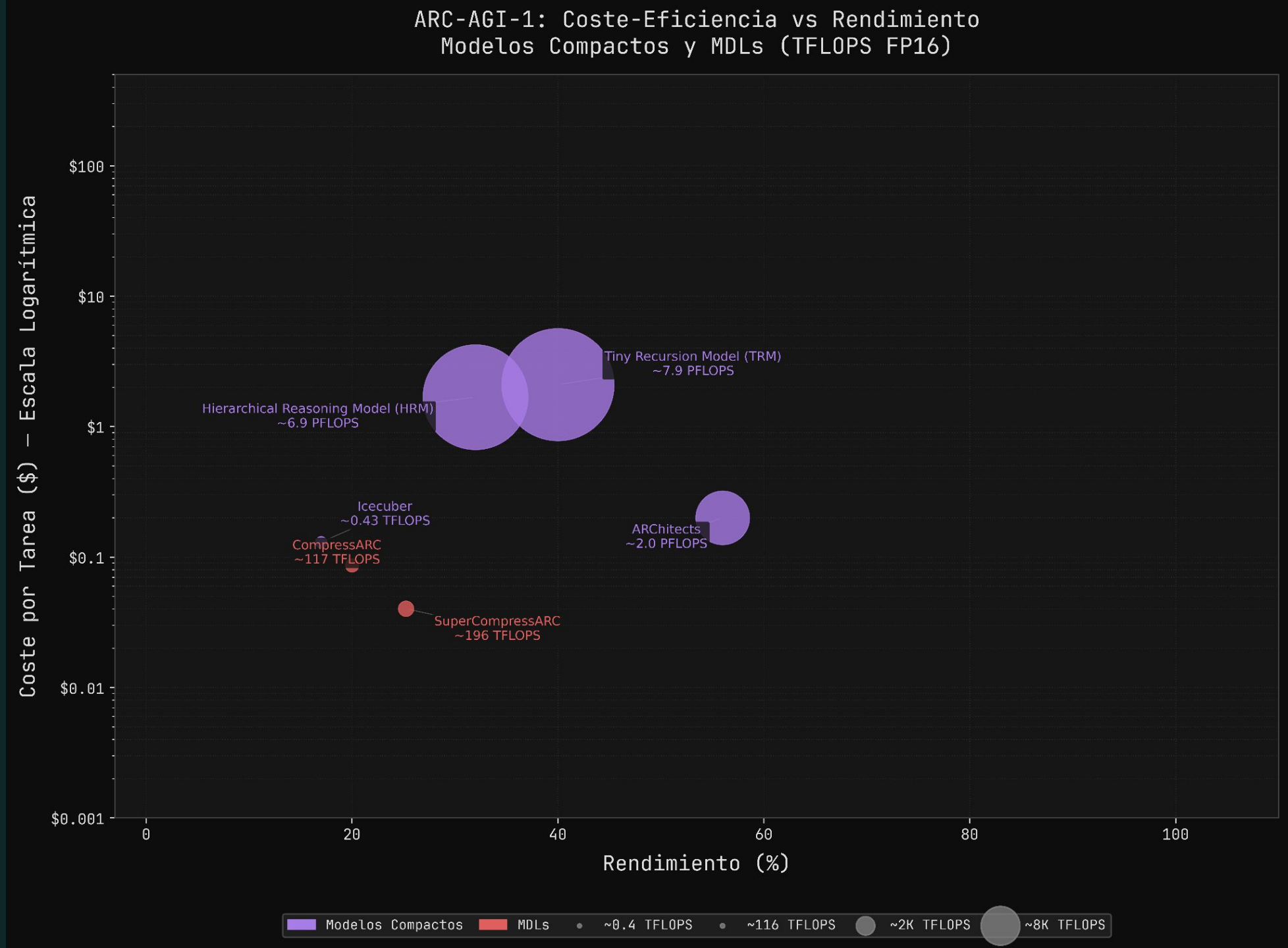

Tiempo total de ejecución (evaluación)
Tiempo (h)
138.0 h
-74.2 %
35.6 h
-53.4 %
64.3 h
Base
Eje H/B
Eje H/B/D

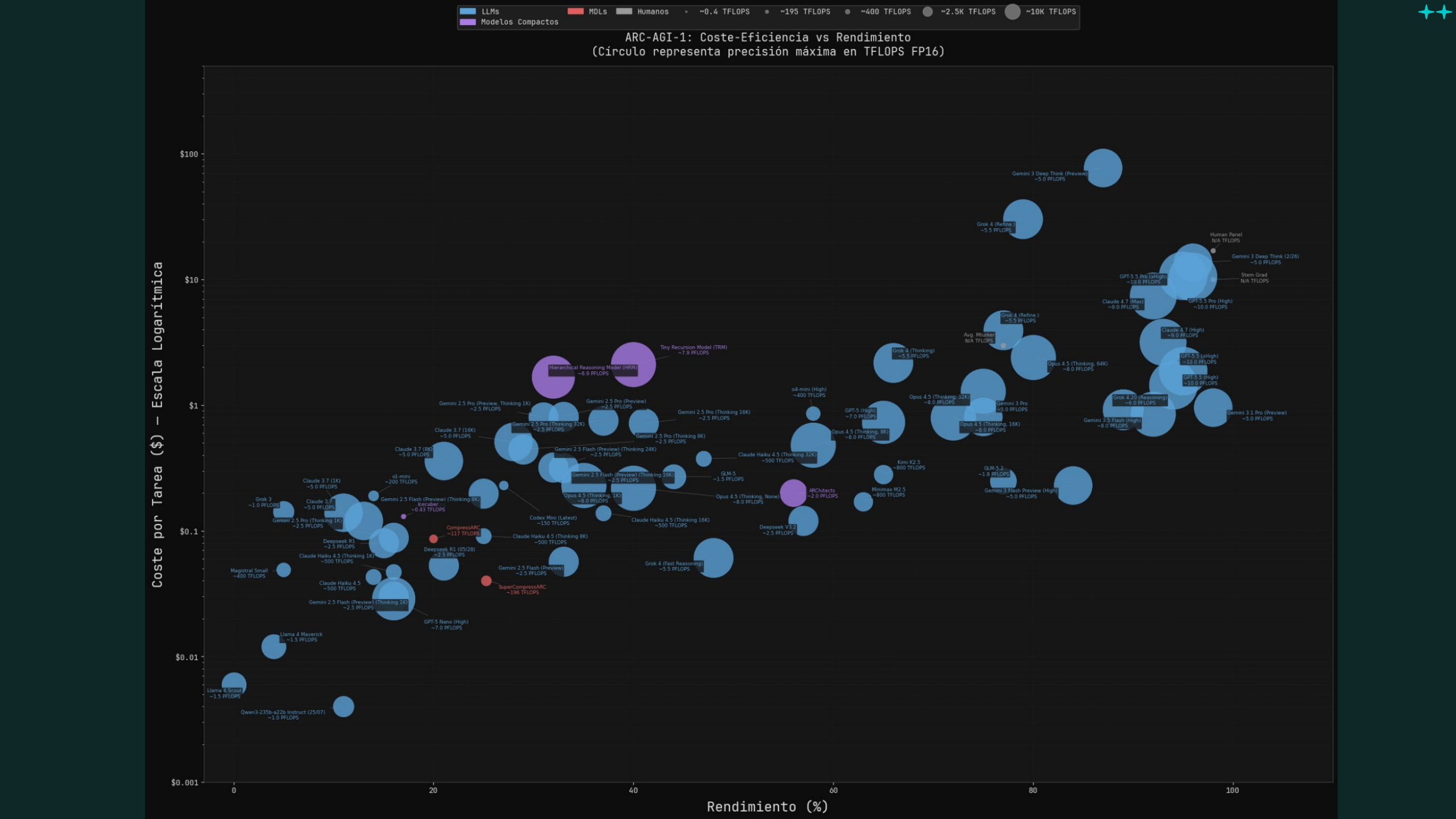

pass@2 Entrenamiento
Tareas resueltas (%)
34.75 %
0.0 %
34.75 %
+26.6 %
44.00 %
Base
Eje H/B
Eje H/B/D

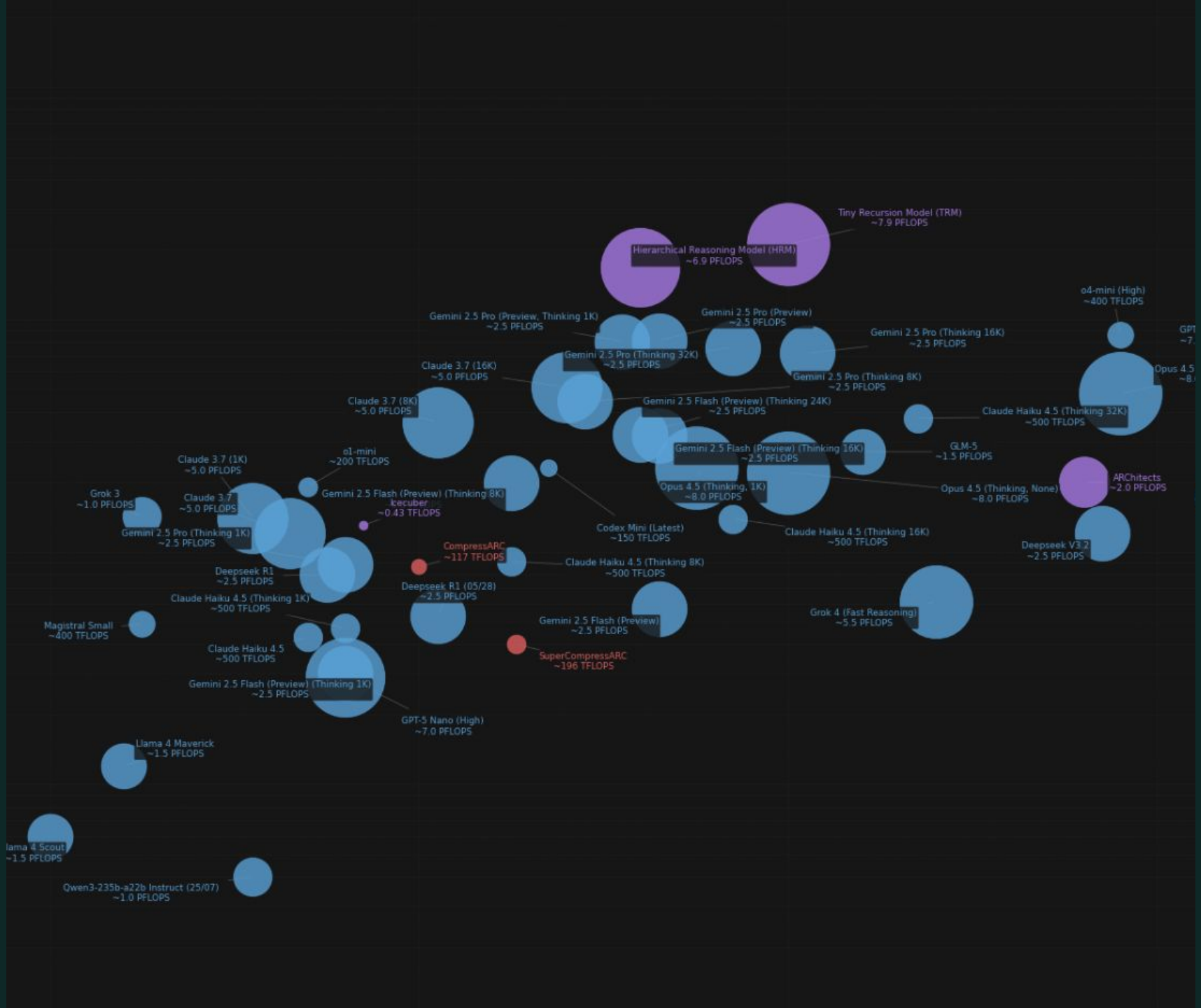

Coste por tarea (aproximación)
Coste por tarea ($)
0.086 $
-74.1 %
0.02225 $
-53.5 %
0.04 $
Base
Eje H/B
Eje H/B/D


pass@2 Evaluación
Tareas resueltas (%)
20.00 %
0.0 %
20.00 %
+26.2 %
25.25 %
Base
Eje H/B
Eje H/B/D


Comparativa de Coste por Tarea para Modelos Compactos
Modelos Compactos
MDLs
Coste por Tarea ($)
$0
$0.5
$1
$1.5
$2
$0.040
$0.086
$0.130
$0.200
$1.680
$2.100
SuperCompressARC
CompressARC
Icecuber
ARChitects
Hierarchical Reasoning Model (HRM)
Tiny Recursion Model (TRM)


ARC-AGI-1: Coste-Eficiencia vs Rendimiento
Modelos Compactos y MDLs (TFLOPS FP16)
Coste por Tarea ($) — Escala Logarítmica
$100
$10
$1
$0.1
$0.01
$0.001
0
20
40
60
80
100
Rendimiento (%)
Tiny Recursion Model (TRM)
~7.9 PFLOPS
Hierarchical Reasoning Model (HRM)
~6.9 PFLOPS
Icecuber
~0.43 TFLOPS
CompressARC
~117 TFLOPS
ARChitects
~2.0 PFLOPS
SuperCompressARC
~196 TFLOPS
Modelos Compactos
MDLs
~0.4 TFLOPS
~116 TFLOPS
~2K TFLOPS
~8K TFLOPS


LLMs
Modelos Compactos
MDLs
Humanos
~0.4 TFLOPS
~195 TFLOPS
~400 TFLOPS
~2.5K TFLOPS
~10K TFLOPS
ARC-AGI-1: Coste-Eficiencia vs Rendimiento
(Círculo representa precisión máxima en TFLOPS FP16)
Coste por Tarea ($) — Escala Logarítmica
$100
$10
$1
$0.1
$0.01
$0.001
0
20
40
60
80
100
Rendimiento (%)
Gemini 3 Deep Think (Preview)
~5.0 PFLOPS
Grok 4 (Refine.)
~5.5 PFLOPS
Human Panel
N/A TFLOPS
Gemini 3 Deep Think (2/26)
~5.0 PFLOPS
GPT-5.5 Pro (xHigh)
~10.0 PFLOPS
Stem Grad
N/A TFLOPS
Claude 4.7 (Max)
~9.0 PFLOPS
GPT-5.5 Pro (High)
~10.0 PFLOPS
Grok 4 (Refine.)
~5.5 PFLOPS
Avg. Mturker
N/A TFLOPS
Claude 4.7 (High)
~9.0 PFLOPS
Opus 4.5 (Thinking, 64K)
~8.0 PFLOPS
GPT-5.5 (xHigh)
~10.0 PFLOPS
GPT-5.5 (High)
~10.0 PFLOPS
Tiny Recursion Model (TRM)
~7.9 PFLOPS
Hierarchical Reasoning Model (HRM)
~6.9 PFLOPS
Grok 4 (Thinking)
~5.5 PFLOPS
o4-mini (High)
~400 TFLOPS
Opus 4.5 (Thinking, 32K)
~8.0 PFLOPS
Gemini 3 Pro
~5.0 PFLOPS
Grok 4.20 (Reasoning)
~6.0 PFLOPS
Gemini 3.1 Pro (Preview)
~5.0 PFLOPS
Gemini 3 Flash (High)
~8.0 PFLOPS
Gemini 2.5 Pro (Preview, Thinking 1K)
~2.5 PFLOPS
Gemini 2.5 Pro (Preview)
~2.5 PFLOPS
Gemini 2.5 Pro (Thinking 16K)
~2.5 PFLOPS
Gemini 2.5 Pro (Thinking 32K)
~2.5 PFLOPS
Gemini 2.5 Pro (Thinking 8K)
~2.5 PFLOPS
Claude 3.7 (16K)
~5.0 PFLOPS
GPT-5 (High)
~7.0 PFLOPS
Opus 4.5 (Thinking, 8K)
~8.0 PFLOPS
Opus 4.5 (Thinking, 16K)
~8.0 PFLOPS
Claude 3.7 (8K)
~5.0 PFLOPS
Gemini 2.5 Flash (Preview) (Thinking 24K)
~2.5 PFLOPS
Claude Haiku 4.5 (Thinking 32K)
~500 TFLOPS
Kimi K2.5
~800 TFLOPS
Claude 3.7 (1K)
~5.0 PFLOPS
o1-mini
~200 TFLOPS
Gemini 2.5 Flash (Preview) (Thinking 16K)
~2.5 PFLOPS
GLM-5
~1.5 PFLOPS
GLM-5.2
~1.8 PFLOPS
Gemini 3 Flash Preview (High)
~5.0 PFLOPS
Grok 3
~1.0 PFLOPS
Claude 3.7
~5.0 PFLOPS
Gemini 2.5 Flash (Preview) (Thinking 8K)
Icecuber
~0.43 TFLOPS
Opus 4.5 (Thinking, 1K)
~8.0 PFLOPS
ARChitects
~2.0 PFLOPS
Opus 4.5 (Thinking, None)
~8.0 PFLOPS
Minimax M2.5
~800 TFLOPS
Gemini 2.5 Pro (Thinking 1K)
~2.5 PFLOPS
Codex Mini (Latest)
~150 TFLOPS
Claude Haiku 4.5 (Thinking 16K)
~500 TFLOPS
Deepseek V3.2
~2.5 PFLOPS
CompressARC
~117 TFLOPS
Claude Haiku 4.5 (Thinking 8K)
~500 TFLOPS
Deepseek R1
~2.5 PFLOPS
Deepseek R1 (05/28)
~2.5 PFLOPS
Claude Haiku 4.5 (Thinking 1K)
~500 TFLOPS
Magistral Small
~400 TFLOPS
Gemini 2.5 Flash (Preview)
~2.5 PFLOPS
Grok 4 (Fast Reasoning)
~5.5 PFLOPS
Claude Haiku 4.5
~500 TFLOPS
SuperCompressARC
~196 TFLOPS
Gemini 2.5 Flash (Preview) (Thinking 1K)
~2.5 PFLOPS
GPT-5 Nano (High)
~7.0 PFLOPS
Llama 4 Maverick
~1.5 PFLOPS
Llama 4 Scout
~1.5 PFLOPS
Qwen3-235b-a22b Instruct (25/07)
~1.0 PFLOPS


Tiny Recursion Model (TRM)
~7.9 PFLOPS
Hierarchical Reasoning Model (HRM)
~6.9 PFLOPS
o4-mini (High)
~400 TFLOPS
Gemini 2.5 Pro (Preview, Thinking 1K)
~2.5 PFLOPS
Gemini 2.5 Pro (Preview)
~2.5 PFLOPS
Gemini 2.5 Pro (Thinking 16K)
~2.5 PFLOPS
Gemini 2.5 Pro (Thinking 32K)
~2.5 PFLOPS
Claude 3.7 (16K)
~5.0 PFLOPS
Gemini 2.5 Pro (Thinking 8K)
~2.5 PFLOPS
Claude 3.7 (8K)
~5.0 PFLOPS
Gemini 2.5 Flash (Preview) (Thinking 24K)
~2.5 PFLOPS
Claude Haiku 4.5 (Thinking 32K)
~500 TFLOPS
Gemini 2.5 Flash (Preview) (Thinking 16K)
~2.5 PFLOPS
GLM-5
~1.5 PFLOPS
Claude 3.7 (1K)
~5.0 PFLOPS
o1-mini
~200 TFLOPS
ARChitects
~2.0 PFLOPS
Grok 3
~1.0 PFLOPS
Claude 3.7
~5.0 PFLOPS
Gemini 2.5 Flash (Preview) (Thinking 8K)
Icecuber
~0.43 TFLOPS
Opus 4.5 (Thinking, 1K)
~8.0 PFLOPS
Opus 4.5 (Thinking, None)
~8.0 PFLOPS
Gemini 2.5 Pro (Thinking 1K)
~2.5 PFLOPS
Codex Mini (Latest)
~150 TFLOPS
Claude Haiku 4.5 (Thinking 16K)
~500 TFLOPS
Deepseek V3.2
~2.5 PFLOPS
CompressARC
~117 TFLOPS
Claude Haiku 4.5 (Thinking 8K)
~500 TFLOPS
Deepseek R1
~2.5 PFLOPS
Deepseek R1 (05/28)
~2.5 PFLOPS
Claude Haiku 4.5 (Thinking 1K)
~500 TFLOPS
Grok 4 (Fast Reasoning)
~5.5 PFLOPS
Magistral Small
~400 TFLOPS
Gemini 2.5 Flash (Preview)
~2.5 PFLOPS
Claude Haiku 4.5
~500 TFLOPS
SuperCompressARC
~196 TFLOPS
Gemini 2.5 Flash (Preview) (Thinking 1K)
~2.5 PFLOPS
GPT-5 Nano (High)
~7.0 PFLOPS
Llama 4 Maverick
~1.5 PFLOPS
lama 4 Scout)
~1.5 PFLOPS
Qwen3-235b-a22b Instruct (25/07)
~1.0 PFLOPS

Conclusiones

- Exploración técnicas de bucles de refinamiento.
- Desarrollar técnicas de entrenamiento en tiempo de inferencia.
- Optimización de consumo de memoria y tiempo de ejecución.
- Comparativa exhaustiva de eficiencia de manera cuantitativa respecto al uso de memoria VRAM y precisión teórica total en FLOPs.

Trabajo futuro

- Mejorar las habilidades primitivas de la arquitectura.

- Rediseño de la arquitectura hacia algo menos transinductivo. Entrenar un modelo con las tareas de entrenamiento y posteriormente usarlo como parte para inferenciar las tareas de evaluación.

- Recurrencia mediante un modelo transformer.

- Usar LoRA para pre-entrenamiento con cambio de paradigma.

¿Preguntas?